

Research Paper

How can urban green space be planned for a ‘happy city’? Evidence from overhead- to eye-level green exposure metrics

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HIGHLIGHTS

- Satellite, land-cover, and street view-derived data were used to assess UGS exposure.
- Multiple UGS exposure metrics are correlated with residents' expressed happiness (EH).
- UGS size shows the highest potential for predicting residents' EH.
- Overhead UGS metrics were more effective in promoting residents' EH than eye-level ones.
- Large, aggregated, and highly vegetated UGSs are prioritized in planning a “happy city”

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ABSTRACT

The benefits of urban green spaces (UGSs) for human health have been extensively documented. Nevertheless, few studies have incorporated multidimensional UGS exposure indicators, and little is known about the effectiveness of different metrics that should be prioritized as nature-based solutions for improving mental well-being. This study aimed to investigate the associations between various UGS exposure metrics and residents' expressed happiness (EH) as well as to determine the prioritization of metrics in Nanjing, China, a megacity with 9.5 million inhabitants. The study region was divided into 500 m × 500 m grids, and 330,000 geotagged posts from social media (*Sina Weibo*) were retrieved for sentiment analysis using the Natural Language Processing (NLP). We developed a systematic UGS exposure framework using satellite, land-cover, and street view-derived data, encompassing 17 indicators of composition and configuration at overhead level as well as street green space visibility and perceived quality at eye level. A regression model and Likelihood Ratio Test were used to examine the associations between various UGS indicators and EH and determine the prioritization of indicators. The results indicated that UGS size had the greatest potential for promoting residents' EH, followed by overall greenness in the surrounding area, aggregated UGS, perceived quality, and visibility of street green spaces. This study also found that overhead-level metrics may be more effective than eye-level metrics in enhancing residents' EH, although both perspectives showed significant associations with EH. These findings provide valuable insights into health-oriented landscapes and urban planning to promote the development of a “happy city,” particularly in low-green resource settings in low- and middle-income countries.

1. Introduction

Rapid urbanization and deficiencies in natural exposure have exacerbated mental health issues in urban residents. Indeed, researchers have engaged in interdisciplinary studies to explore ways to enhance people's mental well-being and what constitutes a “happy city” (Ballas,

2013; De Vries et al., 2021). According to Montgomery (2013), planning for “happy cities” (similar to Green or Low Carbon Cities) requires the active incorporation of climate, energy, resilience, recreation, and other dimensions. As a “health-supported environment,” urban green spaces (UGSs) are nature-based solutions that can provide regulating ecosystem services to protect the ecological integrity of cities and offer cultural

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ecosystem services to promote the mental well-being and happiness of urban residents (Bratman et al., 2019; Dadvand et al., 2016). However, UGSs encompass a diverse array of types and the exposure measurements are multidimensional (e.g., overhead vs. eye level), which might yield various health benefits through multiple mechanisms (Markevych et al., 2017; Marselle et al., 2021). Identifying the key indicators of UGS exposure that effectively and efficiently enhances residents' mental well-being is a crucial step in landscape and urban planning to determine the prioritization of UGS interventions, thereby promoting residents' quality of life, urban sustainability, and the development of a "happy city".

1.1. UGS exposure indicators at overhead level: Composition and configuration

Current research on UGS exposure has primarily focused on overhead-level indicators, incorporating UGS composition and configuration. Most studies have concentrated on UGS composition (i.e., the proportion of different types of UGS elements coverage), which is always measured as 'quantity' in epidemiological studies, such as the normalized difference vegetation index (NDVI), enhanced vegetation index, percentage of tree canopy coverage, and vegetation coverage (Bratman et al., 2019). Several studies also considered UGS configuration, which refers to the spatial character, arrangement, connectivity, or position of patches of UGS, such as size, fragmentation, and shape complexity (Grafius et al., 2018; Sodoudi et al., 2018). Among them, size is a widely used configuration metric for assessing UGS exposure (Schipperijn et al., 2013).

1.2. UGS exposure indicators at eye level: Quantity and perceived quality

Overhead-level UGS exposure metrics are less likely to reflect actual pedestrian interactions with UGSs because of their inability to measure the vertical dimensions of exposure (Jiang et al., 2017); in this regard, recent studies have developed eye-level metrics (e.g., green view index) to objectively measure visible greenery in urban areas using street-view images (e.g., Google Street View) through deep learning techniques (Yao et al., 2021). More recently, several studies have begun to consider the quality of street green space at eye level, utilizing a couple of attractiveness and aesthetic indicators, such as accessibility, walkability, and safety (Rui and Cheng, 2023; Zhang, 2023).

1.3. UGS and mental well-being: Toward multidimensional exposures

Previous studies have used various overhead- and eye-level metrics to assess UGS exposure. However, most cross-sectional and longitudinal epidemiological studies have primarily focused on overhead UGS exposure indicators. Specifically, higher NDVI (Crouse et al., 2021), green areas (Bertram and Rehdanz, 2015), vegetation coverage (Yuan et al., 2018), and tree canopies (Zhang et al., 2022) have consistently been found to have a protective effect on mental well-being. Notably, a few studies have also underscored the importance of UGS configuration in promoting residents' health benefits. Nevertheless, this evidence predominantly concentrates on mortality and physical health (e.g., cardiovascular and respiratory diseases) (Schinas et al., 2023). Limited evidence has established a link between UGS configuration and mental well-being, especially in low- and middle-income countries (LMICs). Moreover, prior research has examined the associations between eye-level green space exposure and mental well-being, with street green space visibility tending to be the most commonly used exposure indicator (Yang et al., 2023). However, few studies have directly examined the association between perceived quality of street green spaces and mental well-being.

Collectively, the greenspace-health literature increasingly advocates for the integration of a variety of exposure metrics, ranging from overhead to eye level and from quantity to quality, to elucidate the nexus

between UGSs and mental well-being (Reyes-Riveros et al., 2021; Wolch et al., 2014; Yu et al., 2024). This shift (from single to multiple exposure dimensions) stems from an expanding recognition that various UGS exposure indicators might influence health outcomes through a spectrum of mechanisms (e.g., mitigation, instoration, and restoration) (Markevych et al., 2017; Marselle et al., 2021). Furthermore, the inherent complexity, diversity, and dynamism in human-nature interactions necessitate a nuanced and comprehensive framework for assessing UGS exposure, which could more effectively disentangle the relationships between UGS and mental well-being. Although a small body of research on the relationship between multidimensional UGS exposure and mental well-being (Liu et al., 2021; Yang et al., 2023), we are unaware of studies that simultaneously consider composition and configuration at overhead level, and quantity and perceived quality at eye level.

1.4. Expressed happiness (EH) index as a representation of mental well-being

As a fundamental component of health and quality of life, mental well-being encompasses aspects of hedonic (pleasure) and eudemonic (happiness) well-being and has been extensively examined in interdisciplinary literature using diverse measures (Ryan and Deci, 2001). Semi-structured interviews and questionnaires are commonly used in self-report measures (Reyes-Riveros et al., 2021; Yuan et al., 2018). These measures are well developed and provide support for literature at the individual level, while their subjectivity and reliance on retrospective data, as well as limited sample sizes, may impact the accuracy of assessments. Moreover, several studies have used smartphone GPS technologies and wearable sensors to obtain real-time happiness data once users contact UGSs (Benita et al., 2019; De Vries et al., 2021). However, these measurements are costly and time-consuming, making it difficult to generalize and replicate assessments on larger scales. Emerging social media data are advantageous because of their broad geographic coverage, large data volumes, and ease of accessibility (Wang et al., 2022a). Recently, these data have been used to indicate the emotions of residents (Roberts et al., 2019). With the development of data crawling and text analysis technologies, sentiment analysis based on natural language processing (NLP) platforms can play an essential role in examining the relationship between environmental exposure and residents' EH (Cheng et al., 2021; Zheng et al., 2019).

1.5. Objectives

This study aimed to investigate the associations between various UGS exposure metrics and residents' EH, and to determine the prioritization of UGS exposure metrics in predicting health outcomes. The assessment of UGS exposure in this study involved a systematic evaluation of a total of 17 indicators, including composition and configuration at the overhead level, and street green space visibility and perceived quality at the eye level. The EH was measured using sentiment analysis techniques based on *Sina Weibo*, the largest open social platform in China (Zheng et al., 2019). We hypothesized that the UGS composition and configuration metrics were significantly associated with residents' EH, as were street green space visibility and perceived quality. No hypothesis regarding the prioritization levels of different exposure metrics was proposed because of the limited evidence surrounding this question. We anticipate the findings emerging from testing these hypotheses to help identify key UGS intervention indicators that promote residents' EH, addressing an urgent need to better understand planning and design strategies that can maximize health benefits and contribute to the creation of a "happy city," particularly in low-green resource settings in LMICs.

2. Materials and methods

2.1. Study site and research design

The study site was the inner-city district of Nanjing, China (Fig. 1). Nanjing is a megacity with an area of 6,596 square kilometers and a total population of ~9.5 million (2022), of which ~8.3 million live in urban areas. The inner-city district is the most prominent area in the city from economic, cultural, and historical perspectives, but it also maintains the highest population density and most complex road networks. Over the past few decades, Nanjing has undergone “green gentrification,” and studies have shown that UGSs tend to be inequitably distributed across the city (Zhang, 2023), with potential health ramifications. Accordingly, analysis of this region is important when studying the association between UGS and EH in urban areas.

The workflow of data processing and analysis is illustrated in Fig. 2. UGS exposure were assessed from two perspectives: overhead and eye levels. Overhead-level metrics assessed UGS composition and configuration, whereas eye-level metrics assessed the street green space visibility and perceived quality. Microblog posts from *Sina Weibo* were obtained to calculate residents’ EH (i.e., served as a measure of mental well-being) using NLP technique. This study site was divided into 500 m × 500 m fishnets (grids) in ArcGIS v.10.7 (Esri, 2011) based on previous literature (e.g., Mazumdar et al., 2021). During the statistical analysis phase, bivariate correlations were examined between residents’ EH and UGS indicators. Subsequently, a regression model was used to explore the associations between various UGS exposure indicators and residents’ EH. The Likelihood Ratio Test (LRT) was then used to evaluate the goodness of fit among models incorporating different UGS exposure dimensions and to determine the prioritization of these UGS metrics in predicting residents’ EH. Finally, sensitivity analyses were conducted to assess the robustness of the findings.

2.2. Assessment of residents’ EH

Weibo (i.e., *Sina Weibo*) is one of China’s top social networking platforms (Wang et al., 2022a), with 511 million monthly active users as of September 2021. It offers geotagged microblog posts in which users agree to share location information (Fig. S1) (Zheng et al., 2019). Geotagged posts from July 2019 to June 2020 were identified using a web crawler based on Python v.3.7 and were connected to a streaming

application programming interface (API). Approximately 330,000 raw posts were obtained, of which approximately 300,000 remained after filtering out invalid data (redundant data, posts with few words, only emoticons, or posts lacking personal opinions).

On social media, individuals express their feelings using words with varying sentiment polarities. To obtain the overall sentiment tendency of the posts and enhance the accuracy of the results by considering the context, sentiment analysis was conducted on all texts of valid posts (each post was analyzed separately) based on the *Baidu* NLP national AI open-innovation platform. After segmenting the Chinese words and extracting the main elements (target, opinion, holder, and time) from the unstructured texts, we used an API provided by the *Baidu* NLP platform to call upon the Chinese sentiment lexicon and analyzed Chinese texts with subjective descriptions in general scenarios. This technique could capture the hedonic state of an individual’s sentiment when a *Weibo* update was posted based on a high-precision computer deep neural network (DNN) semantic model and lexicon (Supplementary S1). The results returned the probability of the positive category, probability of the negative category, sentiment polarity classification results, and confidence level of classification. The probabilities ranged from 0 to 1. According to Wang et al. (2022a), the probabilities of positive and negative sentiment categories can be described as expressed sentiment. We defined the probability of a positive sentiment category as the value of EH (Zheng et al., 2019). The closer the value is to 0, the lower the EH. Specific geolocations of posts with sentiment values were transformed into latitudes and longitudes before visualization in ArcGIS. The number of sentiment points within each grid was then calculated after removing grids with < 30 check-in points to prevent large deviations. The average of all sentiment points within each grid was then calculated as the grid EH value.

2.3. UGS exposure metrics measurement

2.3.1. Overhead-level metrics

The overhead-level metrics of UGS included composition and configuration. In addition to the widely used NDVI, this study used the proportions of tree cover, grassland, and overall UGS to assess the UGS composition. Sentinel-2 satellite imagery from September 2021 (10 m resolution) was acquired from the Copernicus Open Access Hub website. The Sentinel Application Platform (SNAP) and ArcGIS were used to process the images and calculate NDVI values across the study area. The

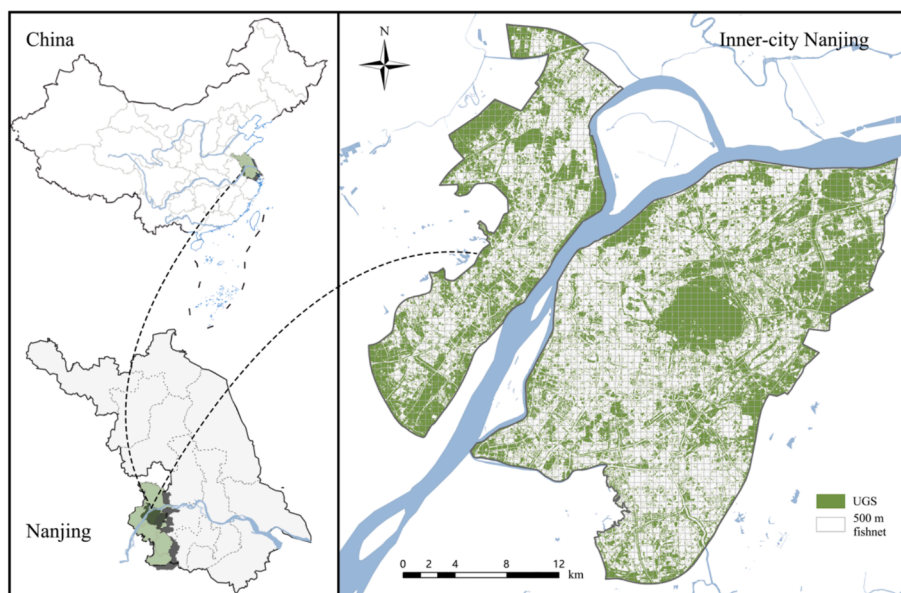


Fig. 1. Schematic figure of the research site and corresponding 500 m × 500 m fishnets.

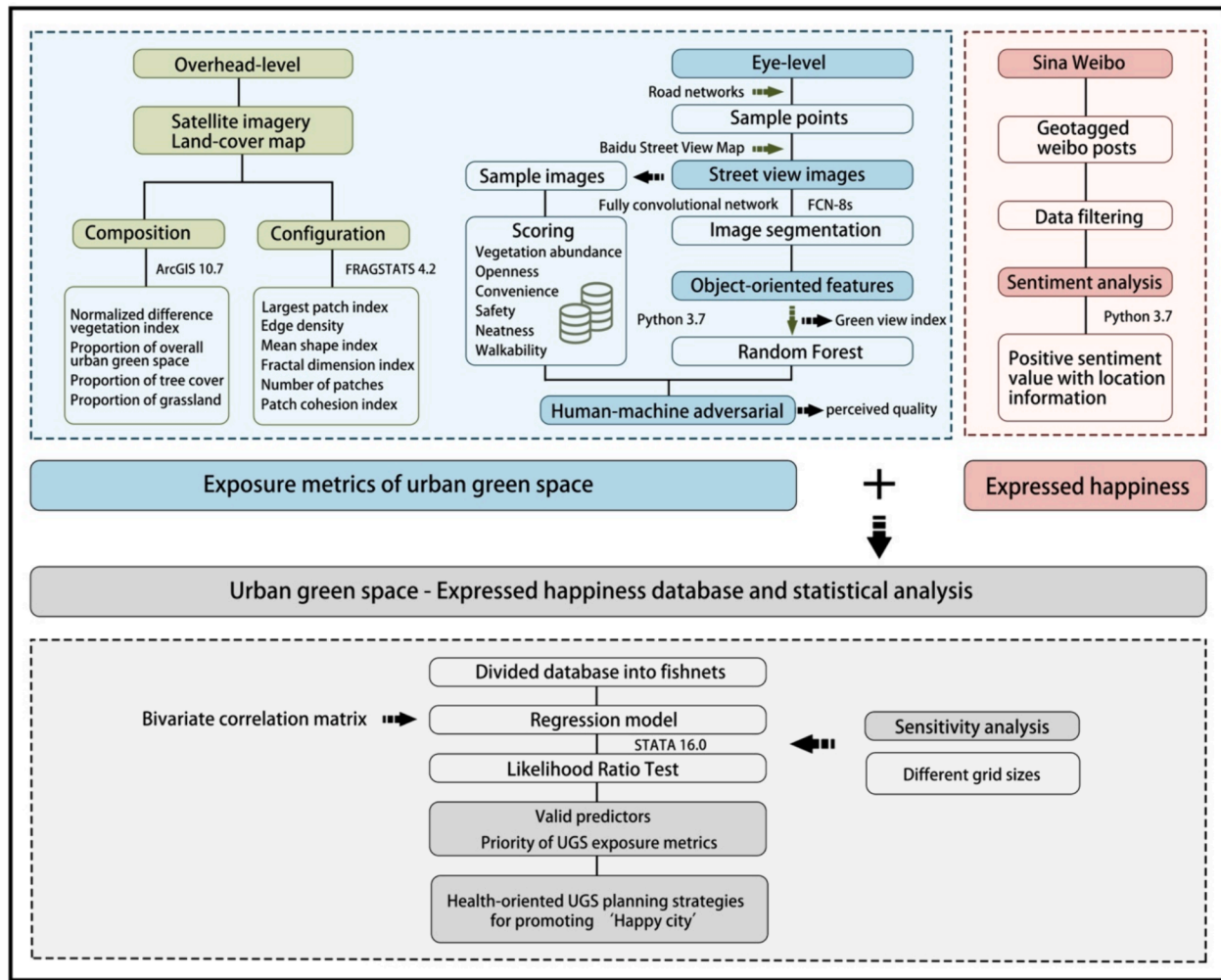


Fig. 2. Workflow of data processing and statistical analysis.

NDVI value of each grid was calculated as the average of all the pixels within the grid, with higher values corresponding to dense vegetation. Negative NDVI values were assigned as “0” to eliminate biases (Rugel et al., 2017). Simultaneously, an image classification process was implemented to produce land-cover maps at 10 m resolution based on Sentinel-2 satellite imagery, which identified seven types of land-cover: tree cover, shrubland, grassland, built-up land, bare land, water bodies, and herbaceous wetlands. Given the small proportion of shrubland (0.001 %) and herbaceous wetlands (0.013 %) in the inner-city district of Nanjing, only tree cover (28.053 %) and grassland (16.090 %) were included as part of the UGS composition metrics.

Subsequently, all types of UGS elements were combined into a single overall UGS metric and imported into FRAGSTATS v.4.2 (McGarigal, 2002) to calculate the UGS structural indices. Six UGS configuration indices were constructed (He et al., 2021), including largest patch index (LPI), edge density (ED), mean shape index (Shape), fractal dimension index (FRAC), number of patches (NP), and patch cohesion index (Cohesion). Among these, LPI measures the UGS size, the closer the value of LPI is to 0, the smaller the largest UGS patch in the grid. Shape and FRAC measure the complexity of UGS shapes, with values closer to 1 indicating simpler shapes (e.g., circles or squares). NP and Cohesion measured the degree of fragmentation and connectedness of the UGS patches, respectively. Low NP and ED values, along with high Cohesion values, indicated relatively aggregated UGS patches, whereas the opposite was true for relatively fragmented UGS patches. Aggregated UGS patches may vary in size and encompass both large and small areas.

2.3.1.1. Eye-level metrics. Street green space visibility and perceived quality metrics were assessed from an eye-level perspective. Specifically, 200 m intervals were established along the road network, and images were captured between the intervals in four cardinal directions (0°, 90°, 180°, and 270°) on the road observation points from Baidu Street View (BSV) using the API (Fig. S3) (Lu et al., 2018). A database of 319,108 BSV images was obtained from 79,777 road observation points across the study region. Significantly, there may be instances where no observation points exist within a 500 m × 500 m grid, or where there is a high concentration of observation points due to the complex road network in the inner city. To address potential biases, grids with fewer than five observation points (i.e., fewer than 20 BSV images) were excluded from this study.

Eye-level metrics were evaluated using the BSV images in two steps: 1) semantic segmentation by the fully convolutional network (FCN-8 s) and 2) residents’ perceptions of BSV images with a human-machine adversarial scoring framework. A DNN model, FCN-8s, was implemented to semantically segment the acquired BSV images and obtain the content composition for each image (Fig. S4). The ADE20K scene parsing and segmentation database, which includes 151 categories (e.g., trees, buildings, cars), was used to train the FCN-8s model (Zhou et al., 2019). Subsequently, the human-machine scoring method was used to obtain residents’ perceptions of street green space in BSV images (Yao et al., 2019). A total of 5,000 images covering various street features were selected from the BSV image database to construct the training dataset. Then, 40 volunteers, including researchers interested in landscape architecture and urban planning as well as those familiar with the

local environment were selected to score the perceived quality indicators of street green spaces from 0 to 100. The male-to-female ratio was 1:1. A random forest model based on Python was used for automatic scoring according to the perceived quality scores provided by the volunteers, along with the proportion of elements corresponding to the semantic segmentation of the images. Hence, a database of eye-level metrics for UGS was obtained, including one street green space visibility metric (Green View Index, GVI) and six perceived quality metrics: vegetation abundance, neatness, safety, openness, convenience, and walkability. The eye-level metric value for each grid was taken as the average of the corresponding metrics for all BSV images. All indicators were standardized to enhance the comparability of the results.

2.4. Covariates

Previous studies have highlighted the potential contributions of environmental factors and socioeconomic indicators on residents' happiness (Brereton et al., 2008; Cheng et al., 2023). To mitigate potential bias in the results, this study incorporated two environmental factors, namely land surface temperature (LST) and the proportion of water bodies, as well as three socioeconomic factors, namely population size, GDP per capita, and housing prices as covariates. LST data were retrieved by applying the radiative transfer equation to the September 2021 Sentinel-2 satellite image in ENVI v.5.3. The proportion of water bodies was calculated from the land-cover map described in Section 2.3.1. We obtained population size data for 2020 from the 100-meter resolution WorldPop database (<https://www.worldpop.org/>). GDP per capita was calculated using real GDP for 2020 at district level and population size data. Housing price data were obtained from LianJia (<http://nj.lianjia.com>), an online housing platform for real estate and rentals in China (Li et al., 2021). All metrics, except the proportion of water bodies, were the average values of the corresponding metrics within each grid. The proportion of water bodies was calculated as the

ratio of the total area of water bodies to the total area of the grid.

2.5. Statistical analysis

The statistical analysis in this study included three steps. Initially, bivariate correlation analyses were performed to examine the relationships between residents' EH and various UGS exposure indicators and to assess the collinearity of the EH predictors. Subsequently, a regression model was applied to each UGS exposure indicator to examine its association with residents' EH and compare the model fits. The base equations for these regression models are given by Eqs. (1):

$$Happiness_i = \alpha_0 + \alpha_1 G_i + \alpha_2 C_i + \varepsilon_0 \quad (1)$$

where $Happiness_i$ represents the residents' EH in grid i , G is the UGS indicator which varies with the studied UGS exposure dimensions, C is the covariate that may affect residents' EH (including LST, the proportion of water bodies, housing price, population, and GDP per capita), α_0 is the constant, and ε_0 is the error term. All regression models conducted were variations of this main model, and performed in STATA v.16.0 (Cameron and Trivedi, 2010), with $p < 0.05$ as the threshold for statistical significance. Higher R-squared and lower Akaike information criterion (AIC) values indicated better model fits. By ranking these values, we identified the metrics that exerted the strongest associations with residents' EH across multiple UGS exposure metrics.

The third step in the analysis involved conducting an LRT for models with UGS indicators at the overhead and eye levels. LRT has been commonly used to assess the goodness of fit between two competing statistical models (nested models) and test whether their difference was statistically significant (Vuong, 1989). A statistically significant difference indicated that the model with more predictors (i.e., complex model) fits the data significantly better than the simple model. Therefore, by comparing the simple model (with the best-fitting indicators at the overhead level) to the complex model (including the best-fitting

Table 1
Summary statistics for each variable in the 635 grids.

Variables	Min.	Max.	Mean	Median	Std.
Dependent variable					
Posts per grid	30	22,958	387	88	1,260
EH	0.534	0.904	0.713	0.713	0.058
Independent variables					
NDVI	0.079	0.775	0.331	0.320	0.105
Overall UGS	0.003	1.000	0.281	0.230	0.200
Tree Cover	0.002	1.000	0.239	0.194	0.190
Grassland	0.000	0.430	0.020	0.004	0.042
LPI	0.236	100.000	18.165	10.179	20.401
ED	0.000	352.935	151.257	150.743	60.838
Shape	1.000	2.834	1.326	1.292	0.170
FRAC	1.003	1.199	1.065	1.064	0.017
NP	1.000	44.000	18.000	18.000	8.000
Cohesion	49.910	100.000	90.437	92.264	7.635
GVI	2.383	75.619	21.056	19.183	10.276
Abundance	8.904	41.280	22.281	21.831	5.883
Neatness	31.944	88.213	68.824	69.555	8.560
Safety	29.440	90.341	64.626	65.189	10.789
Openness	43.028	84.107	62.872	63.273	6.959
Convenience	31.331	86.750	65.727	66.763	8.611
Walkability	32.743	88.500	69.302	71.177	10.174
Covariates					
LST	8.311	29.027	20.007	19.970	3.187
Proportion of water bodies	0.000	0.597	0.021	0.000	0.062
Population size	2	1413	142	63	185
GDP per capita (RMB)	3748.464	35273.590	12241.160	5424.457	10710.800
Housing prices (RMB)	1295.000	91458.170	34992.040	33230.000	13140.270

Note: EH, expressed happiness index; NDVI, normalized difference vegetation index; Overall UGS: Proportion of overall UGS; Tree cover, the proportion of tree cover; Grassland, Proportion of grassland; LPI, largest patch index; ED, edge density; Shape, mean shape index; FRAC, fractal dimension index; NP, number of patches; Cohesion, patch cohesion index; GVI, green view index; LST: Land surface temperature.

indicators from both overhead and eye levels) using LRT, we can prioritize exposure dimensions (i.e., overhead level vs. eye level) in predicting EH.

2.6. Sensitivity analysis

A sensitivity test was conducted to examine the robustness of these results (Velentgas et al., 2013). In addition to the 500 m grid size used in the primary model, grid sizes of 300 m and 1000 m were selected for the sensitivity analyses (Bertram and Rehndanz, 2015; Fuertes et al., 2020). The values of all indicators were calculated according to the corresponding grid sizes, and regression models. Likelihood ratio tests were also re-run for comparison with the main model. Sensitivity analyses were performed using ArcGIS v.10.7 and STATA v.16.0.

3. Results

3.1. Summary statistics

After excluding grids with fewer than 30 check-in points from *Sina Weibo* and fewer than five road observation points, the total number of samples (grids) was 635 (Fig. S2). Table 1 presents the summary statistics for all variables in the sample.

3.2. Results of correlation analyses

Fig. 3 shows the correlations between multiple UGS exposure metrics and EH. Regarding overhead-level metrics, all UGS composition indicators exhibited a positive correlation with residents' EH. The coefficient was observed for overall UGS at 0.230, Tree cover at 0.203, NDVI at 0.193, and Grassland at 0.107. Regarding configuration indicators, NP showed a negative correlation with residents' EH at -0.174 , while LPI, Shape, and Cohesion demonstrated a positive correlation with EH at 0.233, 0.130, and 0.085. Eye-level metrics, such as GVI, Abundance, Safety, and Neatness, were positively correlated with residents' EH at 0.110, 0.090, 0.091, and 0.117.

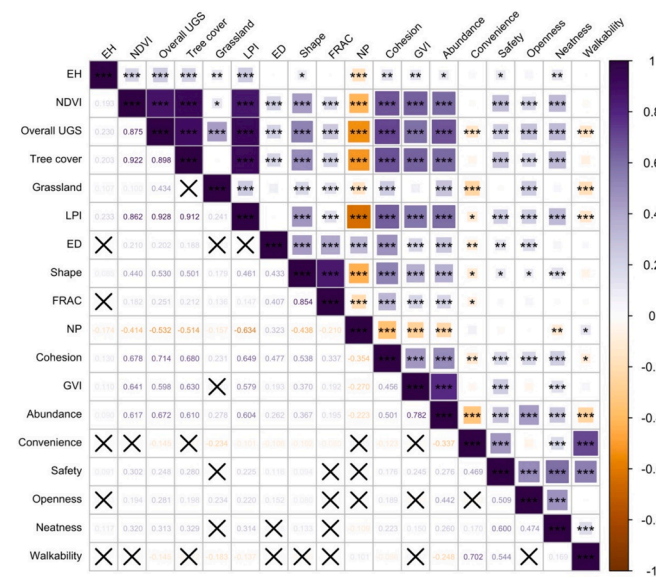


Fig. 3. Correlation matrix between UGS exposure indicators ($N = 635$). **Note:** EH, expressed happiness index; NDVI, normalized difference vegetation index; Overall UGS: Proportion of overall UGS; Tree cover, the proportion of tree cover; Grassland, Proportion of grassland; LPI, largest patch index; ED, edge density; Shape, mean shape index; FRAC, fractal dimension index; NP, number of patches; Cohesion, patch cohesion index; GVI, green view index; LST: Land surface temperature.

3.3. Results of regression analyses

The fitting results for each UGS exposure indicator are presented in Tables 2 and 3 for the overhead and eye levels, respectively. We report the coefficient and p-value of each indicator, as well as the adjusted R-squared and AIC values of each model after adjusting for the built environment and socioeconomic factors. The composition metrics showed significant and positive associations with EH. Among these, the proportion of overall UGS exhibited the best model fit, followed by NDVI, proportion of tree cover, and proportion of grassland. Regarding the configuration indicators, the UGS size indicator (LPI) and Cohesion were positively associated with EH, while NP exhibited significant negative associations with EH. The model fit was superior for LPI, followed by NP and cohesion. Overall, we observed that the UGS size indicator (LPI) explained the greatest variation in the data, followed by the composition indicators (Overall UGS, NDVI, Tree cover, and Grassland) and UGS aggregation indicators (NP and Cohesion).

As shown in Table 3, a one standardized unit increase in GVI was significantly associated with a 0.114 standardized unit increase in residents' EH. Among the seven perceived quality indicators, only Abundance (coefficient = 0.105) and Neatness (coefficient = 0.093) showed significant positive associations with EH, with Abundance having a better model fit. Additionally, we observed that Abundance, the indicator with the largest contribution among the perceived quality indicators, accounted for a comparable level of variation in the data as the quantity indicator.

The preceding analyses enabled us to determine the metrics that had the strongest associations with residents' EH across various dimensions. These metrics were ranked in descending order of impact as follows: configuration-related LPI (R-squared 0.080), composition-related Overall UGS (0.076), perceived quality-related Abundance (0.053), and quantity-related GVI (0.051). Subsequently, an LRT was performed

Table 2

Regression results for different UGS exposure metrics at overhead level.

Overhead-level metrics	Coefficient	P-value	Adj R-squared	AIC
Composition indicators				
NDVI	0.174 (0.094, 0.255)	0.000***	0.068	-607.836
Overall UGS	0.153 (0.091, 0.214)	0.000***	0.076	-613.382
Tree Cover	0.134 (0.069, 0.199)	0.000***	0.065	-606.040
Grassland	0.243 (0.111, 0.375)	0.000***	0.061	-602.870
Configuration indicators				
LPI	0.155 (0.096, 0.215)	0.000***	0.080	-616.100
ED	0.005 (-0.066, 0.076)	0.897	0.041	-589.716
Shape	0.082 (-0.051, 0.215)	0.226	0.043	-591.182
FRAC	-0.045 (-0.185, 0.094)	0.521	0.042	-590.115
NP	-0.115 (-0.183, -0.046)	0.001***	0.057	-600.632
Cohesion	0.106 (0.025, 0.186)	0.010**	0.051	-596.386

Note: The numbers in parentheses indicate the 95 % confidence intervals (CI) of the coefficients; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Adj R-squared, adjusted R-squared; AIC, Akaike Information Criterion; NDVI, normalized difference vegetation index; Overall UGS, proportion of overall UGS; Tree cover, proportion of tree cover; Grassland, proportion of grassland; LPI, largest patch index; ED, edge density; Shape, mean shape index; FRAC, fractal dimension index; NP, number of patches; Cohesion, patch cohesion index. All models included five covariates: land surface temperature (LST), housing price, population, gross domestic product (GDP) per capita, and proportion of water bodies.

Table 3

Regression results for different UGS exposure metrics at eye level.

Eye-level metrics	Coefficient	P-value	Adj R-squared	AIC
Quantity indicator				
GVI	0.114 (0.027, 0.201)	0.011*	0.051	−596.303
Perceived quality indicators				
Abundance	0.105 (0.032, 0.178)	0.005**	0.053	−597.731
Neatness	0.093 (0.009, 0.177)	0.030*	0.048	−594.486
Safety	0.067 (−0.005, 0.139)	0.069	0.046	−593.036
Openness	0.060 (−0.016, 0.136)	0.121	0.045	−592.129
Convenience	0.002 (−0.080, 0.083)	0.971	0.041	−589.700
Walkability	−0.015 (−0.083, 0.053)	0.673	0.041	−589.879

Note: The numbers in parentheses indicate the 95 % confidence intervals (CI) of the coefficients; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Adj R-squared, adjusted R-squared; AIC, Akaike Information Criterion; GVI, green view index. All models included five covariates: land surface temperature (LST), housing price, population, gross domestic product (GDP) per capita, and proportion of water bodies.

to compare the contributions of the overhead- and eye-level indicators using two models (the simple model including LPI, and Overall UGS at overhead level, and the complex model including LPI and Overall UGS at overhead level, as well as GVI and Abundance at eye level). The LRT yielded a test statistic of 2.42 with a p-value of 0.090, exceeding the 0.05 threshold, indicating that the inclusion of eye-level indicators did not significantly enhance the fit of model.

3.4. Results of sensitivity tests

In the sensitivity analysis based on different grid sizes (Table S1-4), the results for the 300 m and 1000 m grids were broadly consistent with the primary 500 m grid model. The results emphasize the positive effects of UGS composition indicators such as NDVI, Overall greenness, and Tree cover, while highlighting the adverse effects of UGS dispersion. Furthermore, a consistent pattern was observed in comparison with the primary model, indicating that the overhead-level UGS metrics were more effective in predicting residents' happiness than eye-level indicators. In summary, the observations from the sensitivity analysis demonstrate the robustness of the study's findings.

4. Discussion

4.1. Prioritization of UGS exposure metrics for promoting residents' EH

This study investigated the association between multidimensional UGS exposure metrics and residents' EH. In doing so, preferable UGS metrics for promoting urban residents' well-being and happiness were identified. These findings suggest that residents living in regions with larger and highly aggregated UGSs (rather than numerous small UGSs), higher proportions of vegetation, and greater street green space visibility and perceived quality tended to be happier than those living in other regions (Fig. 4). The following subsections discuss the preferred UGS exposure metrics for potentially enhancing residents' EH.

4.1.1. Preferred UGS metrics at overhead level

Consistent with previous research (Labib et al., 2020), this study demonstrated the positive associations between UGS composition

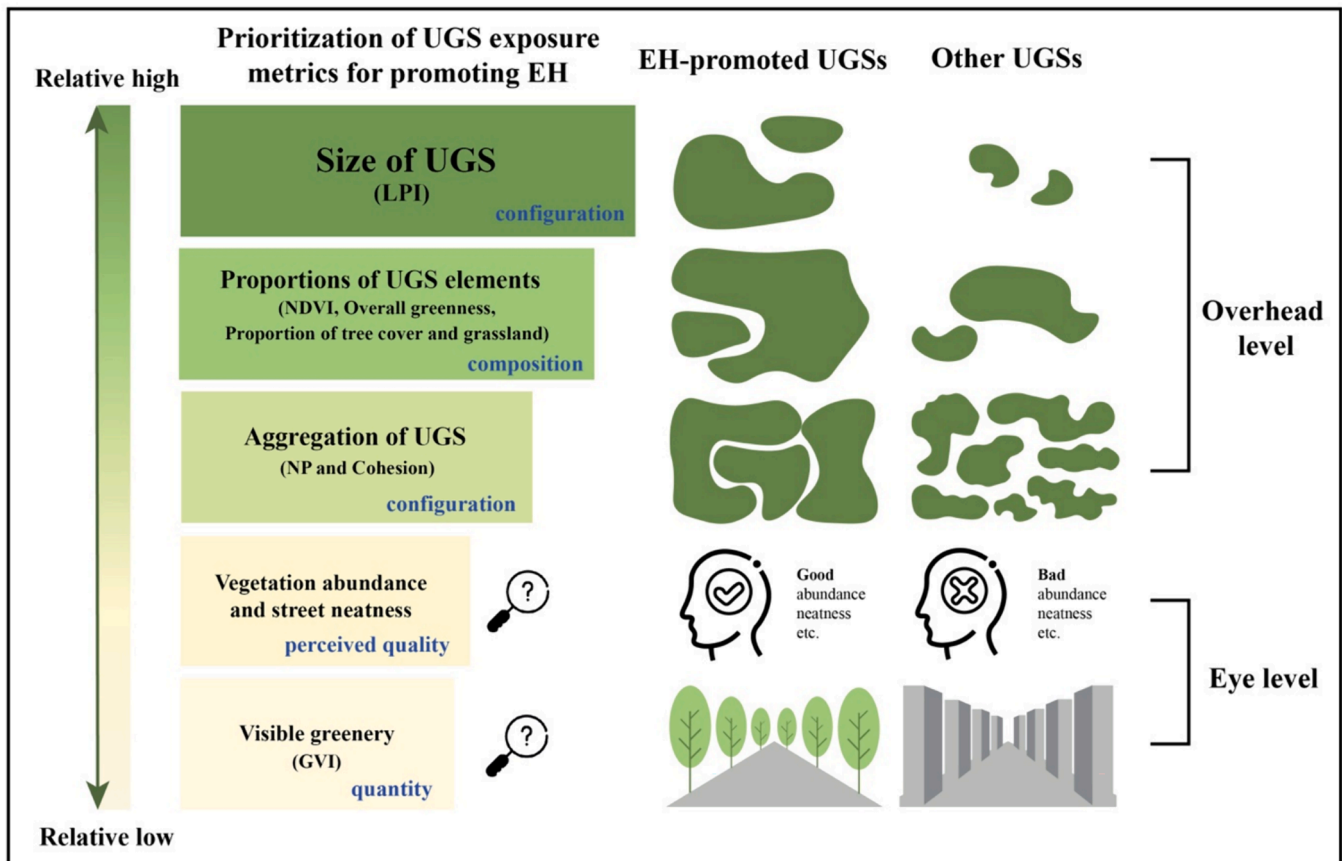


Fig. 4. Visualization of the prioritization for various UGS exposure metrics **Note:** UGS, urban green space; LPI, largest patch index; NDVI, normalized difference vegetation index; NP, number of patches; Cohesion, patch cohesion index; GVI, green view index.

indicators and residents' EH. Specifically, prior studies have shown that surrounding greenness is significantly correlated with happiness (Patino et al., 2023), and have highlighted that higher levels of vegetated greenness may significantly enhance residents' mental well-being (Liu et al., 2021). This study further revealed that the potential contribution of woodlands (i.e., the proportion of tree cover) was more pronounced than that of grasslands, although higher proportions of grasslands also had a positive association with residents' EH. This finding is in line with a cohort study suggesting that increased daily exposure to woodland, rather than grassland, is associated with higher cognitive development scores and a lower risk of emotional problems in adolescents (Maes et al., 2021).

This study also examined the association between UGS configuration and residents' EH. We found that UGS size had the strongest association with residents' EH, which is consistent with a systematic review by Reyes-Riveros et al. (2021), who found that the number of UGS and their size significantly enhanced mental well-being. Similarly, a recent empirical study by Kong et al. (2022) demonstrated that UGS areas were significantly correlated with visitors' positive emotions in Beijing. One possible explanation is that the larger size of UGS may be more capable of offering a diverse range of ecosystem services, which in turn can effectively improve the mental well-being of residents (Bratman et al., 2019). Additionally, we found that the highly aggregated UGSs were more likely to promote residents' EH. However, these findings contrast with those of Ha et al. (2022), who demonstrated that disaggregated UGSs could effectively reduce psychological distress in Chicago. These inconsistencies may be attributed to different study scales and mental health outcomes. Ha et al. (2022) focused on the self-reported mental well-being of residents living within specific neighborhoods, whereas social media data posted by residents anywhere in the city (e.g., homes, workplaces, recreational fields, and shopping centers) were used in this study to assess EH. Given the limited literature, we could not further compare the effects of UGS configuration metrics on mental well-being with those reported in previous epidemiological studies. Nevertheless, several studies have demonstrated the positive associations between highly aggregated UGSs and physical health. For example, He et al. (2021) suggested that larger and more concentrated UGSs may reduce all-cause mortality in certain groups. Similarly, Wang and Tassinari (2019) found that high UGS aggregation reduced the risk of heart disease, chronic lower respiratory disease, and neoplasms.

4.1.2. Preferred UGS metrics at eye-level

A significant role of street green space visibility has been demonstrated in the association between UGS exposure and EH, which is consistent with prior research (Yang et al., 2023). For example, Helbich et al. (2019) highlighted that geriatric depression was mitigated by exposure to street green spaces in Beijing; similarly, Wang et al. (2022b) proposed that higher levels of visible UGSs were associated with lower levels of negative emotions in Guangzhou. Moreover, this study built upon the literature by unraveling the associations between the perceived quality of street green spaces and residents' EH. This broadly aligns with findings from a recent study that investigated the impacts of dynamic greenspace exposure on residents' mental health in Guangzhou, highlighting the vital role of the perceived quality of street green spaces (Wang et al., 2021). This study further addressed the potential contribution of the abundance and neatness of street green space to the improvement of residents' EH. A possible explanation is that well-maintained streetscapes with abundant vegetation have a greater aesthetic appeal and attractiveness, potentially encouraging active commuting, which in turn may enhance the EH of residents (Wu et al., 2020).

Collectively, this study confirmed that the overhead-level metrics were more effective than those at the eye level in enhancing the EH of residents. These findings are also supported by Yue et al. (2022), who argued that UGS exposure indicators measured using overhead views were more strongly associated with mental health than those measured

by street views in Dalian, China. This result may be explained by the overhead-level UGS metrics that more accurately reflect the complex characteristics and multiple types of UGSs, including composition and configuration, whereas the eye-level metrics mainly reflect visual contact with street green spaces.

4.2. Implications of UGS planning for "happy city" development

This study supports the feasibility of investing in UGS as a nature-based solution to promote urban residents' mental well-being (Bratman et al., 2019). Notably, the findings demonstrated that various dimensions of UGS exposure metrics contribute differently to residents' EH, suggesting that urban planners should consider multidimensional aspects when implementing health-oriented UGS planning. This is in line with the recent 3-30-300 rule of thumb for greener, healthier, and more resilient cities proposed by Konijnendijk (2022), which emphasizes the necessity of comprehensively enhancing UGS exposure through visual contact (three visible trees), surroundings (30 % tree canopy coverage), and proximity to recreational UGSs (300 m from the nearest public park) (Browning et al., 2023).

Uniquely, the present study developed a systematic UGS exposure framework, including 17 indicators ranging from UGS composition to configuration from an overhead-level perspective, as well as street green space visibility to perceived quality from an eye-level perspective. This framework can serve as a valuable tool for urban planners to improve UGS exposure for promoting human health. Additionally, this study utilized voluntarily published posts on social media by residents to analyze EH. This strategy considers both the geographical dimension and spatial heterogeneity and can potentially enrich datasets of well-being and happiness in convenient, inexpensive, and replicable ways. By linking residents' EH to multiple UGS exposures, several crucial indicators that offer efficient and effective improvements in residents' EH were identified. Specifically, decision-makers and urban planners should prioritize the provision of larger and aggregated green spaces, while also enhancing the city's overall greenness and the visibility and quality of street green space when implementing health-oriented UGS planning to build a "happy city".

4.3. Limitations

This study has some limitations. First, different types of UGSs (e.g., public parks and gardens) were not considered. Second, although this study suggests several valid metrics for predicting EH in residents, the thresholds for the corresponding indicators remain unknown. Third, a potential bias may arise when utilizing social media to evaluate residents' EH. Social media users trend toward younger generations (Zheng et al., 2019), which may result in demographic data omissions in EH assessments. Additionally, this study did not extract user ID information, thereby limiting our access to individual-level data. The representation of data might be skewed by varying posting frequencies among individuals, potentially affecting the assessment of EH.

5. Conclusion

This study developed a systematic UGS exposure framework, including UGS composition and configuration at the overhead level, as well as street green space visibility and perceived quality at eye level. We evaluated residents' EH using social media data, examined the associations between various UGS exposure metrics and resident EH, and determined the prioritization of these UGS indicators. The results suggested positive associations between UGS indicators and EH at both the overhead and eye levels. UGS size exhibited the strongest association with residents' EH, followed by overall greenness, aggregated UGS, perceived quality, and the visibility of street green spaces. We also observed that overhead-level metrics may be more effective than eye-level metrics in enhancing residents' EH. Such findings could not only

enrich the theoretical understanding of UGS–health relationships but also provide novel insights into enhancing residents' well-being through effective UGS planning, thereby promoting the development of a “happy city”.

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CRediT authorship contribution statement

Yingyi Cheng: Writing – review & editing, Writing – original draft, Formal analysis, Data curation, Conceptualization. **Matthew H.E.M. Browning:** Writing – review & editing, Methodology, Conceptualization. **Bing Zhao:** Funding acquisition, Conceptualization. **Bing Qiu:** Funding acquisition, Conceptualization. **Hengyuan Wang:** Data curation, Conceptualization. **Jinguang Zhang:** Funding acquisition, Conceptualization, Writing – original draft, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.landurbplan.2024.105131>.

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